

Arcs complete outside a prescribed conic

A secant-defect identity, lower bounds, and verified examples

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Abstract

Let $\mathcal{C}$ be a nonsingular conic in $\text{PG}(2, q)$. We study arcs $A \subseteq \text{PG}(2, q) \setminus \mathcal{C}$ whose secants cover every point outside $A \cup \mathcal{C}$, and write $\rho_{\mathcal{C}}(q)$ for their minimum size. Starting from the classical first two equations for the secant-index distribution of an arc, we prove an exact defect identity for an arbitrary prescribed set of points that may remain uncovered. Its conic specialization gives

$$q^2 - k \leq \binom{k}{2}(q - 1) - \frac{6}{\lfloor k/2 \rfloor} \binom{k}{4} - \frac{I_{\mathcal{C}}(A)}{\lfloor k/2 \rfloor}$$

for every $\mathcal{C}$ -complete k -arc, together with an exact equality criterion and a quantitative stability estimate. In particular,

$$\rho_{\mathcal{C}}(q) \geq \sqrt{2q} + \frac{3}{2} - O(q^{-1/2}).$$

Conversely, projective averaging transfers sufficiently small ordinary complete arcs off the prescribed conic, so $\rho_{\mathcal{C}}(q) \leq t_2(2, q)$ whenever $t_2(2, q) \leq q$. We also give tangent and parity constraints in even characteristic. Directly checkable witnesses, combined with the analytic lower bound, yield

$$\rho_{\mathcal{C}}(8) = \rho_{\mathcal{C}}(9) = \rho_{\mathcal{C}}(11) = 6, \quad 8 \leq \rho_{\mathcal{C}}(16) \leq 9.$$

A supplementary verifier enumerates the relevant projective planes and checks every incidence claim used for these finite examples.

1 Introduction

An arc in a projective plane is a set of points no three of which are collinear. An arc is complete when it cannot be enlarged while remaining an arc; equivalently, every point outside it lies on a secant determined by two of its points. Complete arcs and the closely related problem of finding small saturating sets have a substantial literature; see, for example, Ball [1], Kim–Vu [4], and Nagy [5].

Fix a nonsingular conic $\mathcal{C} \subset \text{PG}(2, q)$. We impose the arc condition but allow the points of $\mathcal{C}$ to remain uncovered.

Definition 1.1. An arc $A \subseteq \text{PG}(2, q) \setminus \mathcal{C}$ is *$\mathcal{C}$ -complete* if every point of

$$\text{PG}(2, q) \setminus (A \cup \mathcal{C})$$

lies on a secant of A . Define

$$\rho_{\mathcal{C}}(q) = \min\{|A| : A \text{ is a } \mathcal{C}\text{-complete arc}\}.$$

For an arbitrary arc disjoint from $\mathcal{C}$, its uncovered locus is

$$\mathcal{U}_{\mathcal{C}}(A) = \text{PG}(2, q) \setminus \left(A \cup \mathcal{C} \cup \bigcup_{\{a,b\} \in \binom{A}{2}} ab \right).$$

Thus A is $\mathcal{C}$ -complete exactly when $\mathcal{U}_{\mathcal{C}}(A) = \emptyset$.

Equivalently, A is maximal among arcs in the point set $\text{PG}(2, q) \setminus \mathcal{C}$. All nonsingular conics in $\text{PG}(2, q)$ are projectively equivalent, so the numerical value of $\rho_{\mathcal{C}}(q)$ does not depend on the chosen conic.

The problem differs from two nearby standard notions. A saturating set need not be an arc and has no prescribed exceptional locus. An almost-complete subset of a conic, in the sense of Bartoli–Davydov–Marcugini–Pambianco [2], consists of selected points *on* the conic whose chords cover the complementary points. Here the selected arc is required to avoid the conic.

The point index with respect to an arc, and the first two double-counting equations for its distribution, are classical; compare Hirschfeld [3, Chapter 9] and Ball [1]. The main observation of this note is that, after splitting those equations over a prescribed exceptional set and its complement, the inequalities normally used to control overlap have an exact remainder. This produces:

- (i) an exact prescribed-hole defect identity and equality criterion;
- (ii) a quantitative stability bound for near equality;
- (iii) a strengthened lower bound with additive asymptotic term $3/2$;
- (iv) a projective-averaging upper-bound transfer from ordinary complete arcs;
- (v) characteristic-two nucleus constraints; and
- (vi) small finite-field results whose upper bounds are independently verifiable from explicit coordinates.

A targeted literature search did not locate Definition 1.1 or the exact defect identity below in this form. This observation is not a priority certificate: the proofs are elementary consequences of classical index equations, and terminology involving relative saturation or holes may occur elsewhere.

2 The classical secant equations

Throughout the remainder of the paper, $q \geq 3$ and $k \geq 3$. Let Π be a projective plane of order q , let A be a k -arc in Π , and let $r_A(x)$ be the number of secants of A through $x \in \Pi \setminus A$. We usually write $r(x)$. Put

$$N = \binom{k}{2}, \quad m = \left\lfloor \frac{k}{2} \right\rfloor.$$

Lemma 2.1. *For every $x \notin A$, one has $r(x) \leq m$.*

Proof. Distinct secants through x cannot share an endpoint in A . Indeed, two lines through the same pair x, a coincide. Hence the secants through x use pairwise disjoint pairs of points of A , of which there are at most $\lfloor k/2 \rfloor$. $\square$

Proposition 2.2 (Classical first and second index equations). *For every k -arc A in a projective plane of order q ,*

$$\sum_{x \notin A} r(x) = \binom{k}{2}(q-1), \quad (1)$$

$$\sum_{x \notin A} \binom{r(x)}{2} = 3 \binom{k}{4}. \quad (2)$$

Proof. There are $\binom{k}{2}$ secants. Each contains $q+1$ points, exactly two of them in A , and therefore contributes $q-1$ incidences with points outside A . This proves (1).

For (2), count unordered pairs of distinct secants through a common external point. By Lemma 2.1, their four endpoints in A are distinct. Conversely, a four-subset of A has three partitions into two unordered pairs. Each partition gives two secants, which meet in a unique point because the plane is projective. Their intersection is outside A : otherwise one of the two secants would contain three points of the arc. Thus every four-subset contributes exactly three. $\square$

Only the projective-plane axioms and the common line size $q+1$ enter these proofs. In particular, the second equation need not hold in a general finite linear space where two lines may be disjoint.

3 An exact defect identity for prescribed holes

The next theorem is stated for an arbitrary exceptional point set. The conic enters only in later specializations.

Let $\mathcal{H} \subseteq \Pi \setminus A$ be any set of points allowed to remain uncovered, and define

$$\begin{aligned} V_{\mathcal{H}}(A) &= \Pi \setminus (A \cup \mathcal{H}), \\ \mathcal{X}_{\mathcal{H}}(A) &= \{x \in V_{\mathcal{H}}(A) : r(x) > 0\}, \\ \mathcal{U}_{\mathcal{H}}(A) &= V_{\mathcal{H}}(A) \setminus \mathcal{X}_{\mathcal{H}}(A), \\ I_{\mathcal{H}}(A) &= \sum_{y \in \mathcal{H}} r(y). \end{aligned}$$

Thus $\mathcal{X}_{\mathcal{H}}(A)$ is the covered required locus, and $\mathcal{U}_{\mathcal{H}}(A)$ is the uncovered required locus.

Splitting Proposition 2.2 over $V_{\mathcal{H}}(A)$ and $\mathcal{H}$ gives

$$\sum_{x \in V_{\mathcal{H}}(A)} r(x) = N(q-1) - I_{\mathcal{H}}(A), \quad (3)$$

$$\sum_{x \in V_{\mathcal{H}}(A)} \binom{r(x)}{2} + \sum_{y \in \mathcal{H}} \binom{r(y)}{2} = 3 \binom{k}{4}. \quad (4)$$

Theorem 3.1 (Prescribed-hole defect identity). *Let A be a k -arc in a projective plane of order q , let $\mathcal{H} \subseteq \Pi \setminus A$, and put $m = \lfloor k/2 \rfloor$. Define*

$$\Delta_{\mathcal{H}}(A) = N(q-1) - \frac{6}{m} \binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m} - |\mathcal{X}_{\mathcal{H}}(A)|.$$

Then

$$\boxed{m\Delta_{\mathcal{H}}(A) = \sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} (r(x)-1)(m-r(x)) + \sum_{y \in \mathcal{H}} r(y)(m-r(y)).} \quad (5)$$

In particular, $\Delta_{\mathcal{H}}(A) \geq 0$.

Proof. Only points of $\mathcal{X}_{\mathcal{H}}(A)$ contribute to the sum in (3). Multiplying the definition of $\Delta_{\mathcal{H}}(A)$ by m and using (3) gives

$$m\Delta_{\mathcal{H}}(A) = m \sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} r(x) + (m-1)I_{\mathcal{H}}(A) - m|\mathcal{X}_{\mathcal{H}}(A)| - 6\binom{k}{4}.$$

By (4),

$$6\binom{k}{4} = 2 \sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} \binom{r(x)}{2} + 2 \sum_{y \in \mathcal{H}} \binom{r(y)}{2}.$$

Substitution and termwise simplification yield

$$\begin{aligned} m(r-1) - 2\binom{r}{2} &= (r-1)(m-r), \\ (m-1)r - 2\binom{r}{2} &= r(m-r), \end{aligned}$$

which proves (5). Every term is nonnegative by Lemma 2.1. $\square$

Corollary 3.2 (Coverage and uncovered-locus bounds). *Under the hypotheses of Theorem 3.1,*

$$|\mathcal{X}_{\mathcal{H}}(A)| \leq N(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m}, \quad (6)$$

and

$$|\mathcal{U}_{\mathcal{H}}(A)| \geq |V_{\mathcal{H}}(A)| - N(q-1) + \frac{6}{m}\binom{k}{4} + \frac{I_{\mathcal{H}}(A)}{m}. \quad (7)$$

Equality holds in either inequality if and only if

$$r(x) \in \{1, m\} \quad (x \in \mathcal{X}_{\mathcal{H}}(A)), \quad r(y) \in \{0, m\} \quad (y \in \mathcal{H}).$$

Proof. The two inequalities are respectively $\Delta_{\mathcal{H}}(A) \geq 0$ and its rearrangement using $|\mathcal{U}_{\mathcal{H}}(A)| = |V_{\mathcal{H}}(A)| - |\mathcal{X}_{\mathcal{H}}(A)|$. The equality statement follows from the vanishing of every nonnegative term in (5). $\square$

The exact remainder also quantifies near equality.

Corollary 3.3 (Quantitative stability). *Assume $m \geq 3$ and put*

$$\begin{aligned} M &= \{x \in \mathcal{X}_{\mathcal{H}}(A) : 2 \leq r(x) \leq m-1\}, \\ J &= \{y \in \mathcal{H} : 1 \leq r(y) \leq m-1\}. \end{aligned}$$

Then

$$(m-2)|M| + (m-1)|J| \leq m\Delta_{\mathcal{H}}(A). \quad (8)$$

Proof. For $2 \leq r \leq m-1$, $(r-1)(m-r) \geq m-2$. For $1 \leq r \leq m-1$, $r(m-r) \geq m-1$. Apply these estimates to (5). $\square$

Thus small defect forces almost every multiply covered required point to have the maximum possible multiplicity m , while almost every hole point met by a secant also has multiplicity m . This is the precise stability content behind the equality heuristic.

4 Specialization to a conic

Let $\mathcal{C}$ be a nonsingular conic in $\text{PG}(2, q)$, and let $A \subseteq \text{PG}(2, q) \setminus \mathcal{C}$ be a k -arc. Since $|\mathcal{C}| = q + 1$, there are $q^2 - k$ points outside $A \cup \mathcal{C}$. Write

$$I_{\mathcal{C}}(A) = \sum_{y \in \mathcal{C}} r(y) = \sum_{\ell \text{ an } A\text{-secant}} |\ell \cap \mathcal{C}|.$$

Corollary 4.1 (Conic-loss and uncovered-locus inequalities). *For every such arc,*

$$|\mathcal{U}_{\mathcal{C}}(A)| \geq q^2 - k - \binom{k}{2}(q - 1) + \frac{6}{m} \binom{k}{4} + \frac{I_{\mathcal{C}}(A)}{m}. \quad (9)$$

If A is $\mathcal{C}$ -complete, then

$$\boxed{q^2 - k \leq \binom{k}{2}(q - 1) - \frac{6}{m} \binom{k}{4} - \frac{I_{\mathcal{C}}(A)}{m}.} \quad (10)$$

The first correction in (10) accounts for unavoidable concurrence among secants; the last accounts for secant incidence spent on the exempt conic.

Define

$$\begin{aligned} L_1(q) &= \min \left\{ k \geq 3 : q^2 - k \leq \binom{k}{2}(q - 1) \right\}, \\ L_2(q) &= \min \left\{ k \geq 3 : q^2 - k \leq \binom{k}{2}(q - 1) - \frac{6}{\lfloor k/2 \rfloor} \binom{k}{4} \right\}. \end{aligned}$$

Then

$$\rho_{\mathcal{C}}(q) \geq L_2(q) \geq L_1(q). \quad (11)$$

The definition of $L_2(q)$ involves only integer arithmetic after multiplication by $\lfloor k/2 \rfloor$.

For calculation, the corrected capacity has the parity-dependent forms

$$\binom{k}{2}(q - 1) - \frac{6}{\lfloor k/2 \rfloor} \binom{k}{4} = \begin{cases} \frac{k-1}{2}(k(q-1) - (k-2)(k-3)), & k \text{ even}, \\ \frac{k}{2}((k-1)(q-1) - (k-2)(k-3)), & k \text{ odd}. \end{cases} \quad (12)$$

For comparison, some values are

q	8	9	11	13	16	17	19
$L_1(q)$	5	5	6	6	7	7	7
$L_2(q)$	6	6	6	7	8	8	8

Thus the second-moment correction alone proves, for example, that a $\mathcal{C}$ -complete arc in $\text{PG}(2, 16)$ has at least eight points.

Theorem 4.2 (Asymptotic lower bound). *As q ranges over prime powers,*

$$\rho_{\mathcal{C}}(q) \geq \sqrt{2q} + \frac{3}{2} - O(q^{-1/2}). \quad (13)$$

Consequently,

$$\liminf_{q \rightarrow \infty} (\rho_{\mathcal{C}}(q) - \sqrt{2q}) \geq \frac{3}{2}. \quad (14)$$

Proof. Let k be the size of a $\mathcal{C}$ -complete arc and put $s = \sqrt{2q}$. The elementary incidence inequality $q^2 - k \leq \binom{k}{2}(q - 1)$ is quadratic in k ; solving it and expanding its positive root gives

$$k \geq s + \frac{1}{2} + O(s^{-1}).$$

If $k \geq s + 2$, then (13) is immediate. Otherwise write $k = s + a$, where the preceding elementary bound and $k < s + 2$ make a bounded.

Since $m = \lfloor k/2 \rfloor \leq k/2$,

$$\frac{6}{m} \binom{k}{4} \geq \frac{12}{k} \binom{k}{4} = \frac{(k-1)(k-2)(k-3)}{2}.$$

Dropping the nonnegative conic term in (10) therefore gives the necessary inequality

$$q^2 - k \leq \frac{k-1}{2} (k(q-1) - (k-2)(k-3)).$$

After substituting $q = s^2/2$ and $k = s + a$, the right side minus the left side is

$$\left(\frac{a}{2} - \frac{3}{4}\right) s^3 + O(s^2),$$

uniformly for bounded a . Nonnegativity forces $a \geq 3/2 - O(s^{-1})$, proving (13) and hence (14). $\square$

Remark 4.3 (The scale of the conic term). Every secant meets $\mathcal{C}$ in at most two points, so

$$0 \leq I_{\mathcal{C}}(A) \leq 2 \binom{k}{2}.$$

At the relevant scale $k \asymp \sqrt{q}$, the term $I_{\mathcal{C}}(A)/m$ is therefore $O(\sqrt{q})$, while the universal overlap correction $6 \binom{k}{4}/m$ is $\Theta(q^{3/2})$. Consequently, improving the lower bound on $I_{\mathcal{C}}(A)$ within (10) can be useful for finely balanced finite cases, but cannot improve the additive constant $3/2$ in Theorem 4.2. Any stronger asymptotic result needs an additional mechanism, not merely a spectral estimate for $I_{\mathcal{C}}(A)$.

5 An upper-bound transfer from complete arcs

Let $t_2(2, q)$ denote the smallest size of an ordinary complete arc in $\text{PG}(2, q)$.

Theorem 5.1 (Projective averaging). *If $t_2(2, q) \leq q$, then*

$$\rho_{\mathcal{C}}(q) \leq t_2(2, q). \quad (15)$$

More generally, every complete b -arc with $b \leq q$ has a projective image disjoint from the prescribed conic.

Proof. Let B be a complete b -arc and choose a uniformly random projectivity $g \in \text{PGL}(3, q)$. Point transitivity gives, for every $b \in B$,

$$\Pr(gb \in \mathcal{C}) = \frac{q+1}{q^2+q+1}.$$

Hence

$$\mathbb{E}|gB \cap \mathcal{C}| = \frac{b(q+1)}{q^2+q+1} < 1$$

when $b \leq q$. Since the intersection size is a nonnegative integer, some g satisfies $gB \cap \mathcal{C} = \emptyset$. The image gB remains complete in the whole plane, and is therefore $\mathcal{C}$ -complete. $\square$

Kim and Vu proved that, for all sufficiently large q , every projective plane of order q has a complete arc of size at most $\sqrt{q}(\log q)^c$ for an absolute constant c [4]. This is less than q for large q , so Theorem 5.1 gives the following immediate consequence.

Corollary 5.2. *For some absolute constant c and all sufficiently large prime powers q ,*

$$\rho_{\mathcal{C}}(q) \leq \sqrt{q}(\log q)^c.$$

The transfer does not exploit the conic and is unlikely to be sharp. It nevertheless supplies a uniform upper bound without having to preserve the arc condition during a separate random covering construction.

6 Even characteristic and the nucleus

Assume q is even and let ν be the nucleus of $\mathcal{C}$. The nucleus is outside $\mathcal{C}$, all tangents to $\mathcal{C}$ pass through ν , and every line through ν is one of these $q + 1$ tangents.

Proposition 6.1 (The nucleus belongs to the arc). *Let A be a k -arc disjoint from $\mathcal{C}$ with $\nu \in A$. Exactly $k - 1$ A -secants are tangent to $\mathcal{C}$. In particular,*

$$I_{\mathcal{C}}(A) \geq k - 1, \quad I_{\mathcal{C}}(A) \equiv k - 1 \pmod{2}.$$

If A is $\mathcal{C}$ -complete, then

$$q^2 - k \leq \binom{k}{2}(q - 1) - \frac{6}{m} \binom{k}{4} - \frac{k - 1}{m}. \quad (16)$$

Proof. For every $a \in A \setminus \{\nu\}$, the line νa is tangent to $\mathcal{C}$, giving $k - 1$ tangent secants. No line joining two points of $A \setminus \{\nu\}$ can be tangent: every tangent contains ν , which would place three points of A on one line. Hence these are exactly the tangent A -secants.

Every tangent contributes one to $I_{\mathcal{C}}(A)$, while every other line contributes zero or two. This proves the lower bound and parity statement. Inequality (16) follows from Corollary 4.1. $\square$

Proposition 6.2 (The nucleus lies outside the arc). *Let A be a $\mathcal{C}$ -complete arc with $\nu \notin A$. Then*

$$1 \leq r(\nu) \leq m,$$

the tangent A -secants are exactly the secants through ν , and

$$I_{\mathcal{C}}(A) \equiv r(\nu) \pmod{2}, \quad I_{\mathcal{C}}(A) \geq r(\nu) \geq 1.$$

Consequently,

$$q^2 - k \leq \binom{k}{2}(q - 1) - \frac{6}{m} \binom{k}{4} - \frac{1}{m}. \quad (17)$$

Proof. The nucleus is a required point outside $A \cup \mathcal{C}$, so relative completeness gives $r(\nu) \geq 1$; Lemma 2.1 gives the upper bound. A line is tangent to $\mathcal{C}$ exactly when it passes through ν . Thus the number of tangent A -secants is $r(\nu)$. Reducing the sum of the intersection sizes $|\ell \cap \mathcal{C}|$ modulo two gives $I_{\mathcal{C}}(A) \equiv r(\nu) \pmod{2}$. The remaining claims follow from Corollary 4.1. $\square$

These two cases are useful structural reductions, but Remark 4.3 limits what the incidence term alone can accomplish. For example, at $(q, k) = (16, 8)$ inequality (10) would be contradicted only by $I_{\mathcal{C}}(A) > 268$, whereas the universal upper bound is $I_{\mathcal{C}}(A) \leq 56$. Thus deciding whether $\rho_{\mathcal{C}}(16)$ is eight or nine requires information beyond a lower bound on $I_{\mathcal{C}}(A)$ inserted into the present inequality.

7 Verified finite-field examples

The following proposition uses no exhaustive nonexistence search. Every lower bound follows from $L_2(q)$; every upper bound is supplied by explicit coordinates checked by a short independent verifier.

Proposition 7.1. *For the standard conic*

$$\mathcal{C} : XZ = Y^2$$

in the indicated projective planes,

$$\rho_{\mathcal{C}}(8) = 6, \quad \rho_{\mathcal{C}}(9) = 6, \quad \rho_{\mathcal{C}}(11) = 6, \quad 8 \leq \rho_{\mathcal{C}}(16) \leq 9.$$

Proof. Equation (11) gives

$$L_2(8) = L_2(9) = L_2(11) = 6, \quad L_2(16) = 8.$$

Appendix A lists $\mathcal{C}$ -complete arcs of sizes six, six, six, and nine, respectively. Their verification is described below. $\square$

For reference, the verified incidence statistics are

q	$ A $	$\binom{ A }{2}$	$q^2 - A $	$I_{\mathcal{C}}(A)$	$\max_{x \notin A} r(x)$
8	6	15	58	16	3
9	6	15	75	13	3
11	6	15	115	0	3
16	9	36	247	32	4

The supplementary Python program `verify_relative_conic_arcs.py` performs exact finite-field arithmetic and, for each witness:

- (a) enumerates and normalizes all $q^2 + q + 1$ projective points;
- (b) checks that $XZ = Y^2$ contains exactly $q + 1$ points;
- (c) verifies disjointness from the conic and that every pair-line contains exactly two selected points;
- (d) verifies that every point outside $A \cup \mathcal{C}$ lies on a selected pair-line; and
- (e) independently checks both equations in Proposition 2.2.

Its SHA-256 digest is

e9508958d604e68c6c3d09fd3afadfaa8a3126508a51f1dfa993e7a7aed5d36a.

The verifier is deliberately simpler than a search program: it certifies the listed upper-bound witnesses but makes no claim about search completeness.

8 Further questions

The elementary theory leaves several concrete problems.

Problem 8.1. Determine $\rho_{\mathcal{C}}(16)$. Proposition 7.1 leaves only the possibilities eight and nine.

Problem 8.2. Construct $\mathcal{C}$ -complete arcs of size $O(\sqrt{q})$, or prove that this is impossible for an infinite family. Theorem 5.1 currently supplies only the general complete-arc bound $O(\sqrt{q}(\log q)^c)$.

Problem 8.3. Classify equality and bounded-defect configurations in Theorem 3.1. The equality pattern requires all nontrivial secant concurrence to occur at points of maximum possible index, a severe combinatorial restriction.

Problem 8.4. Determine whether the conic stabilizer or its coherent configuration yields useful finite-order restrictions not reducible to a bound on $I_{\mathcal{C}}(A)$. Remark 4.3 shows that an estimate for $I_{\mathcal{C}}(A)$ alone cannot improve the asymptotic constant obtained here.

The central distinction is between universal overlap, already measured by the classical second index equation, and genuinely conic-specific geometry. The former gives the lower bound in this paper. Progress beyond it will probably require an explicit construction, an algebraic obstruction, or a classification of near-extremal secant arrangements rather than another application of the same incidence budget.

A Witness coordinates and field encodings

All coordinates are homogeneous and use the standard conic $XZ = Y^2$.

For $q = 8$, write $\mathbb{F}_8 = \mathbb{F}_2[\alpha]/(\alpha^3 + \alpha + 1)$ and encode $a_0 + a_1\alpha + a_2\alpha^2$ by the binary integer $a_0 + 2a_1 + 4a_2$. A witness is

$$(0, 1, 1), (0, 1, 2), (1, 0, 1), \\ (1, 0, 2), (1, 1, 2), (1, 1, 6).$$

For $q = 9$, write $\mathbb{F}_9 = \mathbb{F}_3[\alpha]/(\alpha^2 + 1)$ and encode $a_0 + a_1\alpha$ by $a_0 + 3a_1$. A witness is

$$(1, 0, 4), (1, 0, 5), (1, 1, 0), \\ (1, 1, 2), (1, 2, 3), (1, 2, 4).$$

For $q = 11$, coordinates lie in the prime field. A witness is

$$(1, 10, 0), (1, 9, 1), (1, 4, 7), \\ (1, 8, 5), (0, 1, 4), (1, 1, 7).$$

For $q = 16$, write $\mathbb{F}_{16} = \mathbb{F}_2[\alpha]/(\alpha^4 + \alpha + 1)$ and use binary coefficient encoding. A witness of size nine is

$$(1, 5, 14), (1, 5, 3), (1, 9, 2), (1, 0, 9), (1, 13, 13), \\ (1, 6, 10), (1, 3, 9), (0, 1, 11), (1, 10, 2).$$

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